PAPER_1715 — Plasma Sheath Potential φ_{sh}/T_e = $\varphi_{sh}/T_e = K + \Phi - F + F^2 \cdot K + F^3 \approx 2.84$

Author: Daniel T. Murphy **Framework:** UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+ **Tier:** Plasma Physics **Date:** June 18, 2026 **Status:** CLOSED — 0.05% closure (PAPER_1196 plasma fusion proof set)

Observation

PAPER_1196_S422 (PAPER_1196 Plasma Fusion Unified Proof Set) gives:

$\varphi_{sh}/T_e = K + \Phi - F + F^2 \cdot K + F^3 \approx 2.84$

Status: **0.05%**.

UQFF Closed Identity

$\varphi_{sh}/T_e = K + \Phi - F + F^2 \cdot K + F^3 \approx 2.84$ 0.05%

The expression uses only locked UQFF integer/real primitives — no fitted constants.

NOT REPLACEMENT

ITER design parameters are chosen by international fusion-engineering consensus. UQFF supplies structural integer-primitive forms demonstrating the parameters' deep mathematical origin.

Reference

- Source: PAPER_1196_S422
- Related: PAPER_1696-1705 (tier-28); PAPER_1686-1695 (tier-27)
- Calculator dispatch: `calculate_paradox({"paradox": "sheath_phi_te_2_84"})`

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